

Microplastic Polymer Analysis Report

Report ID	657e1737-9cbd-499e-926b-8532467f9b5a
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Sample ID	FTIR-4308

Prediction Results

Detected Polymer	HDPE
Confidence Score	0.94%
Model Used	finetuned_model

FTIR Spectral Analysis

- 2915 cm^{-1} → CH_2 asymmetric stretching
- 1470 cm^{-1} → CH_2 bending

Confidence Interpretation

■■■■■■■■■■ 0.94%

Scientific References

[1] Harper, C. Handbook of Plastics Technologies.

[2] Thompson, R. Plastic debris in oceans.

[3] Barnes, D. Accumulation of plastics.

[4] Andrady, A. Environmental impacts of plastics.

[5] Jung, M. FTIR analysis of HDPE.

AI Generated Summary

The analyzed sample was identified as HDPE with a confidence score of 0.94%. The analysis indicates a reliable classification suitable for environmental microplastic identification.